When Proxy-Selected Checkpoints Hide Reliability Failures under Shift

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Abstract

Checkpoint selection is part of evaluation: the reported model is only as reliable as the rule that chose it. We study objectives that downweight high-loss examples and show that their objective-aligned validation proxy can select checkpoints that look attractive on the proxy, and sometimes on mean accuracy, while worsening held-out loss, calibration, and tail or group reliability under shift. In Camelyon17 ERM, proxy-only selection under a P95 soft-clipping stress test worsens held-out loss and ECE on $10/10$ seeds, with a fixed teacher-tail stress diagnostic worsening as supporting evidence. In end-to-end Camelyon17 finetuning, proxy selection raises mean held-out accuracy from 0.683 to 0.708 while worsening held-out loss from 0.902 to 1.520, validation-side ECE from 0.0308 to 0.0522, and the same diagnostic from 7.25 to 13.48. On frozen CivilComments, a baseline-calibrated validation-loss veto changes the run-level verdict, vetoing all $10/10$ suppressive trajectories admitted by proxy selection. We scope the claim carefully: most evidence compares full reporting workflows, where objective, trajectory, and selector change together; selector-only conclusions come only from labeled same-trajectory selector or admissibility checks. Hard bootstrapping, generalized cross-entropy, teacher-free loss-tail checks, and objective-family controls show that the warning is not unique to soft clipping or to the teacher-tail diagnostic. We recommend a reporting rule, not a new optimizer: report both proxy-selected and validation-loss-selected checkpoints, and treat harmful disagreement on held-out reliability as an unvalidated reliability claim.

1 Introduction

Under distribution shift, evaluation depends not only on the metrics reported but also on the checkpoint selected for reporting. Prior work shows that model-selection rules and stopping criteria can materially change out-of-distribution conclusions [15, 6, 2, 36, 33, 25, 4]. We study this as an evaluation-practice failure mode: when a shaped objective induces its own validation proxy, what reliability claim does the proxy-selected report support under shift?

Many modern pipelines optimize a shaped objective rather than plain cross-entropy or raw MSE [22, 29, 37]. We focus on one subset, which we call *suppressive*: objectives or target transforms that reduce gradient pressure on high-loss examples. Soft clipping is a controlled classification example, and winsorized targets in heavy-tailed regression are a natural regression example. Once chosen, such an objective induces its own validation score, which we call the *proxy*. The resulting workflow is simple: train with the shaped objective, pick the checkpoint with the best validation proxy, and report headline held-out metrics.

Our question is whether that workflow remains trustworthy under shift. For suppressive objectives, it often does not. The proxy-favored checkpoint can look best on the proxy, and in some cases even

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on held-out accuracy, while doing worse on held-out loss, calibration, and group- or tail-sensitive reliability.

Scope. Our claim is specific. We do not claim that validation-loss selection is a universal selector, or that every shaped objective creates the same failure. Most evidence is a baseline-referenced workflow comparison: the objective, trajectory, and selector change together. Selector-only conclusions are limited to explicitly labeled same-trajectory selector and admissibility checks. The largest mean accuracy-up / reliability-down pattern appears on Camelyon17 finetuning, where proxy-only selection raises mean held-out accuracy while worsening held-out loss, validation-side calibration, and a teacher-tail stress diagnostic. The teacher-tail diagnostic is a fixed held-out bank of examples that a reference model finds hard; we use it as a stress check, not as a semantic or clinical subgroup. The primary reliability evidence is held-out loss, calibration, and, where available, official group-sensitive metrics.

The expected part is that suppressing high-loss examples can hurt tail-sensitive metrics. The evaluation hazard is that scalarized checkpoint selection can hide that damage and legitimize the checkpoint that caused it.

The core anchors are Camelyon17 ERM, Camelyon17 finetuning, and frozen CivilComments. CelebA and ACSIncome serve narrower scope checks: CelebA shows transience, and ACSIncome tests the regression analogue. Additional GCE, hard-bootstrap, teacher-free tail, averaging, objective-family, and time-course analyses test mechanism and scope rather than expanding the main claim.

Prior work already established that selector choice matters under shift [15, 6, 2, 36, 33, 25, 4]. Our narrower contribution is to show a specific reporting failure: suppressive proxy-selected workflows can report checkpoints that look attractive on the proxy or on accuracy while failing standard held-out reliability checks. Within-trajectory selector disagreement is the sharpest evidence, and baseline-referenced comparisons show what a suppressive pipeline can present as a reported win. The rest of the paper tests when this hazard persists, when a validation-loss diagnostic helps, and whether the risky regime is tied to suppression rather than to objective shaping broadly.

Contributions. As an evaluation-design contribution, the paper makes three claims.

- **Reporting failure.** Checkpoint selection is part of the evaluation protocol: under objectives that downweight high-loss examples, proxy-selected reporting can make less reliable checkpoints look successful under shift.
- **Scope separation.** We separate workflow comparisons, where objective, trajectory, and selector change together, from same-trajectory selector/admissibility checks. The main evidence uses held-out loss, calibration, and official WILDS worst-group loss, with teacher-tail used as supporting fixed-bank stress evidence.
- **Reporting rule, not optimizer.** Report both proxy-selected and validation-loss-selected checkpoints, and treat harmful disagreement on held-out reliability metrics as an unvalidated reliability claim.

2 Related Work

Our framing is closest to work showing that model selection under shift is itself an evaluation-design problem. DomainBed foregrounds careful tuning and fair comparison in domain generalization [15], SWAD and Ensemble of Averages modify averaging and stopping behavior to stabilize out-of-domain selection [6, 2], Change is Hard studies how selector choice changes subgroup-robustness conclusions [36], ERM++ strengthens the ERM baseline partly through more disciplined selection choices [33], and recent OOD work studies how stopping rules affect both generalization and calibration [25, 4]. Checkpoint averaging, stronger stopping rules, and label-free selector diagnostics are complementary rather than direct substitutes. These works mainly propose or evaluate alternative return rules; our object of study is different: we audit what reliability claim remains valid when a shaped training workflow reports the checkpoint selected by its own proxy. Appendix Table 21 adds a deliberately narrow same-trajectory averaging sensitivity on our SoftClip and GCE runs; it is not a SWAD or Ensemble-of-Averages benchmark, but checks whether simple averaging removes the proxy-selected reliability gap. We therefore do not claim validation-loss selection is a universal selector; the reporting protocol is to expose proxy-selected and validation-loss-selected checkpoints before making loss, calibration, or tail/group reliability claims under shift. Underspecification gives a nearby framing at the pipeline level: predictors that look equivalent on the in-distribution validation surface can still diverge materially at deployment [7]. What is different here is not the generic claim that selection matters. It is the narrower protocol claim that, under suppressive objectives, a run can become misleading when selected by its own proxy, that validation loss can still fail on a cross-modal anchor, and that the resulting selector mismatch can be transient or persistent depending on how suppressive orientation evolves over training. In that sense, underspecification is about deployment-divergent models returned by a pipeline, while this paper studies underspecification of the *selection protocol* inside a suppressive training workflow.

The paper also connects to proxy misspecification and Goodhart-style overoptimization, where optimizing a measurable proxy ceases to track the quantity that actually matters [24, 11, 20, 21]. We study the same basic failure mode in a supervised-learning setting. The issue is not reward optimization at inference time, but checkpoint selection under shaped training objectives and distribution shift. What is selected as the "best" checkpoint can therefore be an artifact of the evaluation protocol itself.

Cross-entropy is a strictly proper scoring rule, while shaped objectives can distort that relationship [32, 13, 30]. This matters here because the selected-checkpoint failures are not only about higher held-out loss; they also show systematic calibration degradation under shift, linking our selector hazard to a broader reliability literature on calibration and uncertainty under distribution shift [16, 27, 19, 5, 8].

We also build on the distribution-shift and robust-learning literature around GroupDRO, domain generalization, shortcut learning, and WILDS benchmarks [31, 18, 1, 15, 12, 17, 3]. Our goal is different from proposing a new robust optimizer. Instead, we ask whether the checkpoint preferred by a suppressive objective remains trustworthy when judged by standard held-out loss and tail-sensitive or group-sensitive diagnostics.

Finally, we connect to work on reshaping hard-example gradients, including focal loss and noisy-label objectives [22, 26, 28, 29, 37]. Our same-stack family controls show that these families are not interchangeable for evaluation: the sign of tail movement differs across families even on the same Camelyon17 ERM stack, and the strongest loss, calibration, and tail degradation concentrate in the suppressive family.

3 Setup

We study a controlled suppressive modification of cross-entropy. For per-example loss $\ell(x, y; \theta)$, threshold τ, and slope $\alpha \in (0, 1]$, the soft-clipped objective is

$$\tilde{\ell}(x, y; \theta) = \begin{cases} \ell(x, y; \theta), & \ell \le \tau, \\ \tau + \alpha(\ell - \tau), & \ell > \tau. \end{cases} \tag{1}$$

Our main experiments use $\alpha = 0.10$ and a fixed P95 threshold calibrated once per stack from the epoch-1 train-loss quantile of the matched baseline run; higher clip thresholds appear only as appendix support. We use soft clipping only as a controlled suppressive perturbation, not as a proposed training recipe.

Selection problem. For a training run that produces checkpoints $\{\theta_t\}_{t=1}^{T}$, let $P_{\mathrm{val}}(\theta_t)$ denote the validation proxy induced by the training objective. The standard proxy-only selector is

$$\hat{\theta}_{\mathrm{proxy}} = \arg\min_t P_{\mathrm{val}}(\theta_t). \tag{2}$$

The baseline checkpoint $\hat{\theta}_{\mathrm{base}}$ is defined analogously under the undistorted objective.

Units of comparison. We use two comparison units throughout the paper. *Baseline-referenced* comparisons compare the suppressive proxy-selected checkpoint with the baseline-selected checkpoint from the undistorted run. *Within-trajectory* comparisons compare alternative selectors on the same suppressive run. Only the second comparison isolates selector choice alone.

Claim type	What varies	Supports
Baseline-referenced workflow hazard	objective, trajectory, selector	workflow-level hazard under suppressive reporting pipelines; not selector-only causality
Within-trajectory selector disagreement	selector / admissibility rule on one suppressive run	literal same-trajectory verdict reversals; not broad selector universality

Reliability metrics. We evaluate checkpoints by held-out loss on the deployment split (CelebA worst-group loss, Camelyon17 hospital-2 loss, CivilComments overall and official WILDS worst-group loss, ACSIncome raw MSE) and by a tail-sensitive diagnostic (vision teacher-difficulty $\mathrm{CVaR}_{0.1}$, ACSIncome tail MSE). For the vision suites, teacher-difficulty $\mathrm{CVaR}_{0.1}$ is computed on two fixed teacher-difficulty evaluation banks (A/B), each with $K = 64$ cells and minimum cell size 20, and then averaged across banks, so the tail comparison is not redefined per checkpoint. For ACSIncome, tail MSE is raw MSE on held-out examples above the training-set P95 target threshold. We refer to this vision metric as the teacher-tail stress diagnostic. It is a diagnostic stress-test metric for the tested vision stacks rather than a universal tail-risk definition. It is not intended to identify clinical subgroups or replace WILDS-style group metrics; it tests whether proxy selection shifts risk into a fixed teacher-defined hard-example slice. On the vision anchors, held-out loss and calibration carry the primary reliability claim; the teacher-tail stress diagnostic is supporting fixed-bank evidence. Held-out accuracy is reported but not treated as sufficient evidence of reliability. Calibration is reported through ECE and Brier score; ECE uses 15 confidence bins. All selectors use validation-side quantities only (proxy, validation loss, admissibility budget), while held-out loss, calibration, and final tail/group metrics are report-only.

Workflow hazard. We call a case a *baseline-referenced workflow hazard* when the suppressive proxy-selected checkpoint improves the proxy relative to baseline but worsens the suite's held-out reliability checks. On the vision anchors, the primary checks are held-out loss and calibration, while the teacher-tail stress diagnostic is supporting fixed-bank evidence; on CivilComments, the reliability check is official WILDS worst-group loss; on ACSIncome, it is tail MSE. We call the hazard *transient* if a matched fixed-horizon comparison no longer shows simultaneous loss and tail worsening, and *persistent* if it does. For CivilComments, where the cross-modal support case is carried by official WILDS worst-group metrics, we use the same matched fixed-horizon comparison with overall-loss and worst-group degradation as the corresponding persistence check. Within-trajectory selector disagreement is assessed separately by comparing proxy-only, validation-loss-only, and diagnostic-veto selection on the same suppressive trajectory.

Validation-loss veto diagnostic. As a simple diagnostic rather than a full solution, we define a budgeted selector, which we call a validation-loss veto (guardrail), that only admits checkpoints whose validation loss stays within a multiplicative budget of the baseline-selected checkpoint:

$$\hat{\theta}_{\mathrm{guard}}(\rho) = \arg\min_{t} P_{\mathrm{val}}(\theta_t) \;\; \text{s.t.} \;\; L_{\mathrm{val}}(\theta_t) \leq \rho \, L_{\mathrm{val}}(\hat{\theta}_{\mathrm{base}}). \tag{3}$$

If no checkpoint from the suppressive run satisfies the budget, we fall back to the baseline checkpoint. We report a small budget sweep $\rho \in \{1.10, 1.25, 1.50\}$ and treat $\rho = 1.25$ as an empirical operating point rather than a derived constant. When fallback happens, we report the suppressive-run admissibility verdict separately from the fallback performance so the veto is read as a diagnostic rather than as a same-trajectory selector win. When checkpoints are already saved, dual reporting adds no retraining cost: select the proxy-best and validation-loss-best checkpoints from the same trajectory, report both on held-out loss, calibration, and the relevant tail/group metric, and use the baseline-calibrated veto only when a trusted undistorted reference run exists. If no trusted undistorted reference run exists, the baseline-calibrated veto should be marked unavailable rather than retrofitted; the safe recommendation is then dual reporting of proxy-best and validation-loss-best checkpoints. If the reference run is itself suppressive, the budget is only relative to that suppressive reference and should not be read as a standard-risk cap against an undistorted baseline. Appendix A.9 also reports an adaptive threshold that uses only baseline validation-loss fluctuations.

Compact mechanistic formalization. Appendix A.1 records the short formal results that organize the empirical findings. First, suppressive weighting creates a non-empty local lure region in which the proxy still improves while standard risk worsens, and stronger suppression widens that region. Second, the matched fixed-horizon split is persistent when early suppressive harm exceeds later recovery and transient when later recovery offsets it. The appendix also records the baseline-referenced loss cap behind the diagnostic veto. These results are descriptive bridges for the empirical analysis, not a general theory of shaped objectives.

Mechanism statistic. To compare objective families on a common Camelyon17 ERM stack, we use the stabilized tail/core gradient-amplification ratio

$$R_w = \frac{\mathbb{E}[g_i^{\mathrm{obj}} \mid \mathrm{tail}]/\mathbb{E}[g_i^{\mathrm{CE}} \mid \mathrm{tail}]}{\mathbb{E}[g_i^{\mathrm{obj}} \mid \mathrm{core}]/\mathbb{E}[g_i^{\mathrm{CE}} \mid \mathrm{core}]}, \tag{4}$$

where g_i is scalar per-example logit-gradient magnitude; tail/core are the top decile of the fixed teacher-difficulty validation bank on the Camelyon17 family-control stack and its complement. Thus $R_w < 1$ indicates relative tail suppression and $R_w > 1$ indicates relative tail upweighting. We use R_w as an operational same-stack warning statistic, not as a theorem about all shaped objectives or a causal proof by itself. Practically, $R_w < 1$ early in training is treated as a warning that proxy-only checkpoint selection may be misaligned with downstream reliability criteria on the tested family-control stack.

4 Experiments

We focus the main paper on the smallest set of settings needed to establish the scientific case.

Datasets and held-out metrics. Camelyon17 is the primary persistent anchor under hospital shift, evaluated with held-out hospital-2 loss, accuracy, calibration, and the teacher-tail stress diagnostic. CelebA is the transient subgroup-shift contrast, CivilComments is the clearest non-vision selector-verdict reversal under overall loss and official WILDS worst-group loss, and ACSIncome is the geographic-shift regression extension with held-out MSE and tail MSE. Because hospital 2 is the held-out deployment domain in Camelyon17, held-out hospital-2 loss is the primary reliability view and coincides with worst-domain loss there. ACSIncome trains on Census divisions $\{1, 2, 3, 5, 7, 9\}$, validates on $\{4, 8\}$, and tests on $\{6\}$ under geographic shift.

Training families. The main paper uses two core training stacks: ERM on frozen embeddings and end-to-end Camelyon17 finetuning. CivilComments uses frozen DistilBERT embeddings with linear probes in the main text, while the strongest end-to-end anchor is tested on Camelyon17. Table 2 keeps the absolute Camelyon17 selector comparison, including the baseline reference row, and adds compact selector-verdict rows for frozen CivilComments and ACSIncome. Appendix A.11 adds a ten-seed end-to-end CivilComments corroboration on an official-group-aware validation surface, but Camelyon17 remains the cleaner end-to-end anchor. Supporting adaptive runs are deferred to the appendix because the paper's main claim is selection-level, not optimizer-level.

Main stress-test setting. We use soft-clipped P95 as the main suppressive stress-test setting. Appendix dose-response sweeps show that the selected-checkpoint harm weakens at P97/P99 but does not disappear immediately on the strongest Camelyon anchors. Higher clip thresholds are retained as supportive appendix sweep detail rather than as co-equal main-text settings.

Selected versus fixed checkpoints. Each suite reports two views. The main view uses the actual proxy-selected checkpoint. The robustness view compares against a matched fixed horizon: epoch 30 for frozen-embedding ERM and epoch 10 for end-to-end finetuning. This matters because the CelebA ERM result is transient while the Camelyon17 cases are persistent.

Sample sizes and controls. The core ERM, objective-family, and finetuning sweeps use $n = 10$ seeds. The objective-family control section compares soft clipping, GCE $q = 0.7$, focal loss [22], and label smoothing on the same Camelyon17 ERM stack, with hard-bootstrap, teacher-free held-out loss-tail, checkpoint-averaging, and end-to-end Camelyon17 finetune controls summarized in the appendix. Additional sweep detail, uncertainty summaries, and supporting adaptive controls are moved to the appendix.

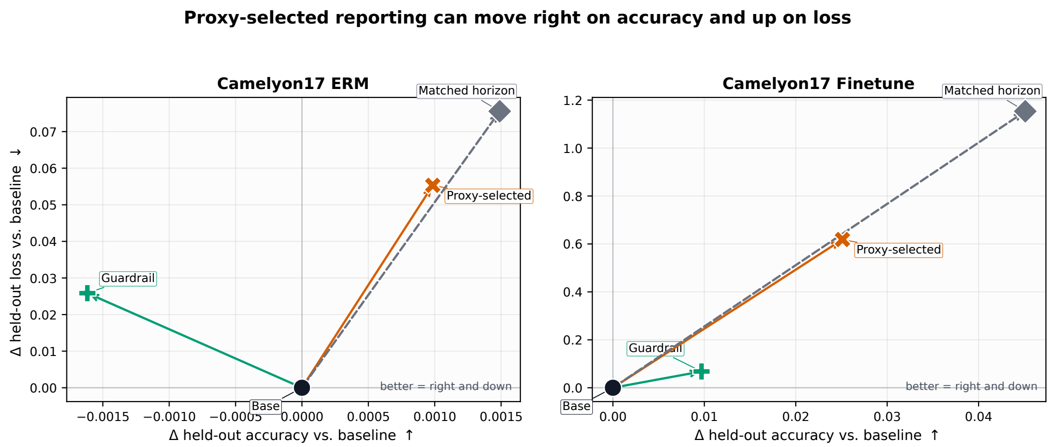

Figure 1: Camelyon17 anchor suites under baseline-referenced and within-trajectory selector views. Points are mean deltas from the baseline-selected checkpoint. Proxy-only, val-loss, and diagnostic-veto selections are evaluated on the suppressive P95 trajectory; the veto may fall back to baseline when no suppressive checkpoint is admissible. In both suites, proxy-only moves up on held-out loss; on finetuning, it also moves right on mean held-out accuracy. Calibration is a primary reliability check in the tables, while the plotted teacher-tail movement is a fixed-bank stress diagnostic. The dashed fixed-horizon arrow tests whether the baseline-referenced harm is only an early-stopping artifact.

5 Results

We organize the evidence at three levels. Baseline-referenced workflow hazards show what a suppressive proxy-selected pipeline can report; same-trajectory selector/admissibility checks isolate checkpoint-return rules on fixed suppressive trajectories; and scope/mechanism controls test whether the pattern is specific to SoftClip or to the teacher-tail diagnostic. This separation is the evaluation-design contribution: a reporting protocol for reliability claims made by proxy-selected checkpoints under shift, not a new optimizer or universal selector. The main evidence comes from Camelyon17 ERM, Camelyon17 finetuning, and frozen CivilComments; ACSIncome is a narrower regression support case, and the remaining analyses are supporting evidence.

5.1 Core Evidence: Proxy-Selected Reporting Hides Reliability Failures

Camelyon17 ERM is the cleanest anchor. Across $10/10$ seeds, proxy-only selection improves the clipped proxy but worsens held-out loss ($+0.055$), ECE ($+0.032$), and the teacher-tail stress diagnostic ($+5.50$). Mean Brier also worsens ($+0.003$), though only on $8/10$ seeds (Appendix Table 16). The paired 95% intervals remain fully on the harmful side for loss ($[+0.045, +0.066]$), tail ($[+5.00, +6.04]$), and ECE ($[+0.030, +0.033]$). A matched epoch-30 check preserves the sign, so the result is not a pure early-stopping artifact. Figure 1 shows the effect visually, Table 1 summarizes the main deltas, and the appendix gives the seed-matched counts and paired intervals.

5.2 Selector and Veto Checks Change the Reliability Verdict

Table 2 makes the comparison semantics explicit. The top panel reports absolute held-out metrics for selectors applied to the two Camelyon17 trajectories. The bottom panel reports delta-from-baseline selector verdicts for frozen CivilComments and ACSIncome. The reported-policy column tells the reader where each row's metrics come from: the baseline run, the same suppressive run, a mixed rule that keeps the suppressive checkpoint when admitted and falls back otherwise, or a pure baseline fallback when no suppressive checkpoint is admissible. The suppressive-admit column records that diagnostic verdict separately.

On the Camelyon anchors, validation-loss-only is safer than proxy-only, but it does not recover the baseline. Five same-trajectory diagnostics support the practical warning. Appendix Table 17

Case	Hazard	ΔAcc↑	ΔLoss↓	ΔTail/rel. diag.↓	ΔECE↓
Camelyon17 ERM	persistent	+0.001	+0.055	+5.501	+0.032
CelebA ERM	transient	-0.017	+2.207	+11.317	+0.033
Camelyon17 Finetune	persistent†	**+0.025**	**+0.619**	**+6.226**	**+0.021**
CivilComments	persistent	-0.004	+0.104	+1.829	+0.070
ACSIncome	persistent	—	+0.027	+0.376	—

Table 1: Core workflow-hazard cases at P95. All deltas are baseline-referenced against the baseline-selected checkpoint from the undistorted run. Vision rows use held-out loss and calibration as primary reliability checks; ECE is held-out for the ERM anchors and validation-side for the Camelyon17 finetune anchor, matching the saved calibration artifacts. The tail/reliability diagnostic column is a teacher-tail stress diagnostic. CivilComments reports official WILDS worst-group loss, and ACSIncome reports tail MSE. CelebA ERM is transient; all other rows are persistent, with † marking a qualitative fixed-horizon tail check for Camelyon17 finetuning. Baseline held-out losses, for scale, are 0.211 (Camelyon17 ERM), 1.557 (CelebA ERM), 0.902 (Camelyon17 finetune), and 0.258 (CivilComments). Every row also satisfies proxy improvement relative to baseline. Appendix Tables 15–16 give seed counts and paired intervals for the classification anchors; the ACSIncome $10/10$ count is backed by the selector artifact. Bold marks the strongest "accuracy up, reliability down" pattern.

shows that the Camelyon signal is visible in standard metrics, not only in the teacher-tail diagnostic. Appendix Table 19 varies the SoftClip threshold: P95/P97 give clean loss/ECE/tail harm relative to validation-loss selection, while P99 attenuates. Appendix Table 20 shows that validation-accuracy selection gives the same warning. Appendix Table 21 adds a same-trajectory selector audit on Camelyon17 ERM: for both SoftClip P95 and GCE ($q = 0.7$), proxy-best is worse than validation-loss-best on held-out loss, ECE, and the teacher-tail stress diagnostic in $10/10$ seeds; last-5 averaging remains harmful, while validation-loss-neighborhood averaging mitigates loss/ECE and leaves tail movement neutral. Appendix Table 22 repeats the standard-metric loss direction across all five leave-one-hospital-out folds. The main SoftClip finetune same-trajectory contrast is mixed, so that headline remains workflow-level; Appendix Table 18 adds a separate end-to-end GCE ($q = 0.7$) selector audit where proxy-best improves the validation proxy and validation accuracy but worsens held-out loss and a fixed teacher-free held-out loss-tail relative to validation-loss-best, with held-out accuracy and calibration mixed. The $1.25\times$ veto is therefore only an empirical admissibility diagnostic; when no comparable non-suppressive baseline run is available, the safer operational recommendation is dual reporting of proxy-selected and validation-loss-selected checkpoints.

Camelyon17 finetuning gives the largest mean accuracy-up / reliability-down pattern. Proxy-only selection raises mean held-out accuracy from 0.683 to 0.708, but held-out loss worsens from 0.902 to 1.520, the teacher-tail stress diagnostic rises from 7.25 to 13.48, and validation-side ECE rises from 0.0308 to 0.0522. The paired intervals remain harmful on loss ($[+0.331, +0.926]$), tail ($[+3.49, +9.07]$), and validation-side ECE ($[+0.018, +0.025]$). So the main claim in this anchor is a baseline-referenced workflow hazard: reliability degrades under proxy-selected suppressive finetuning, not selector-only causality. The accuracy increase is descriptive rather than interval-stable, and the fixed-horizon tail check is qualitative only.

On frozen CivilComments, proxy-only and validation-loss-only give the same harmful standard-metric verdict in the bundled summary. The reversal appears only when a baseline-calibrated validation-loss admissibility rule vetoes the suppressive run and falls back to baseline. This is the paper's clearest baseline-calibrated run-level admissibility reversal on a fixed suppressive trajectory. The paired intervals remain harmful on overall loss ($[+0.100, +0.107]$), official worst-group loss ($[+1.785, +1.871]$), and ECE ($[+0.068, +0.071]$).

On ACSIncome, validation-loss-only reduces the winsorization damage ($+0.023$ raw MSE, $+0.282$ tail MSE), but the $1.25\times$ validation-loss veto is slack and coincides with proxy-only. This is why we treat the veto as a diagnostic rather than a universal selector. We treat ACSIncome as a narrower regression support case rather than as a co-equal anchor with the two Camelyon suites and frozen CivilComments. Appendix Table 10 gives paired intervals for the proxy-only $\rightarrow$ veto reduction on the two Camelyon anchors, and those intervals remain positive for loss and tail reduction at the shown operating point.

Suite	Selector / rule	Reported policy	Supp. admit	Sel. epoch	Acc / ΔAcc	Loss / ΔLoss	Rel. / ΔRel
Camelyon17 selector comparison (absolute held-out metrics across baseline and suppressive trajectories)							
Camelyon17 ERM	Baseline	baseline run	—	5.5	0.919	0.211	5.80
	Proxy-only	suppressive run	1.00	17.0	0.920	0.266	11.31
	Val-loss	suppressive run	1.00	5.3	0.917	0.246	9.30
	1.25x veto	mixed report: admit or fallback	0.70	5.5	0.917	0.237	8.32
Camelyon17 Finetune	Baseline	baseline run	—	1.6	0.683	0.902	7.25
	Proxy-only	suppressive run	1.00	3.9	0.708	1.520	13.48
	Val-loss	suppressive run	1.00	1.8	0.677	1.511	10.05
	1.25x veto	mixed report: admit or fallback	0.40	1.4	0.693	0.970	7.71
Cross-modal / regression selector verdicts (deltas from baseline)							
Frozen CivilComments	Proxy-only	suppressive run	1.00	—	-0.004	$+0.104$	$+1.829$
	Val-loss	suppressive run	1.00	—	-0.004	$+0.104$	$+1.829$
	1.25x veto	baseline fallback	0.00	—	$+0.000$	$+0.000$	$+0.000$
ACSIncome	Proxy-only	suppressive run	1.00	—	—	$+0.027$	$+0.376$
	Val-loss	suppressive run	1.00	—	—	$+0.023$	$+0.282$
	1.25x veto	suppressive run	1.00	—	—	$+0.027$	$+0.376$

Table 2: Selector comparison with semantics made explicit. Top panel: absolute held-out metrics for the two Camelyon17 trajectories; bottom panel: delta-from-baseline verdicts for frozen CivilComments and ACSIncome. Selection epochs are shown only for the Camelyon absolute checkpoint rows; the frozen CivilComments and ACSIncome rows are compact delta-from-baseline verdict summaries. Reported policy identifies whether metrics come from the baseline run, the same suppressive trajectory, an admit-or-fallback mixed report, or pure baseline fallback. Suppressive-admit is the diagnostic verdict for the suppressive run; baseline rows are not applicable. Frozen CivilComments changes from admitted to vetoed under the validation-loss admissibility diagnostic; on ACSIncome, validation loss is milder but the $1.25\times$ veto is slack.

5.3 When the Failure Persists and When It Does Not

CelebA ERM shows that not every hazard is persistent: worst-group loss increases by $+2.21$ and the teacher-tail stress diagnostic by $+11.32$ under proxy-only selection, but fixed epoch-30 deltas reverse sign (-0.70 and -5.93). Camelyon17 ERM remains harmful at the fixed horizon, so that case is not just an early-stopping artifact. Camelyon17 finetuning also keeps harmful fixed-horizon loss, though the fixed-horizon mean loss increase is smaller than at the selected checkpoint ($+0.491$ versus $+0.619$). Its fixed-horizon tail result should be read only qualitatively because a few seeds explode at epoch 10 (Appendix Table 8). Appendix Table 8 gives the selected-versus-fixed comparison for every row still labeled persistent, including frozen CivilComments and ACSIncome at its matched final epoch. Appendix Table 4 and Figure 3 give the supporting mechanism bridge: CelebA's suppressive orientation relaxes ($\Delta R_w = +0.102$), while Camelyon17's entrenches ($\Delta R_w = -0.091$), so CelebA is a transient counterexample rather than a contradiction.

5.4 The Failure Is Not Unique to Soft Clipping

Two ten-seed noisy-label-objective checks on the same Camelyon17 ERM stack address the concern that the result is specific to soft clipping at P95. Generalized cross-entropy (GCE) with $q = 0.7$ gives the strongest corroboration: it is suppressive on this stack at the selected checkpoint ($R_w = 0.339$), but it is a standard noisy-label objective rather than a hand-designed clipping stress test. On the same GCE trajectories, proxy-best selects later checkpoints than validation-loss selection (mean epoch 14.4 versus 3.3), lowers the GCE validation proxy by 0.0146 (95% CI $[-0.0175, -0.0116]$), and raises held-out accuracy by $+0.0098$ ($[+0.0076, +0.0118]$), yet worsens held-out hospital-2 loss by $+0.0181$ ($[+0.0136, +0.0233]$), ECE by $+0.0129$ ($[+0.0104, +0.0153]$), and teacher-tail by $+3.05$ ($[+2.50, +3.60]$) on $10/10$ seeds (Appendix Table 25). Baseline-referenced and fixed-epoch-30 GCE contrasts preserve the accuracy-up / reliability-down direction, so this corroboration is not specific to controlled soft clipping. Hard bootstrapping gives a smaller second noisy-label check: relative to val-loss-only selection inside the same bootstrap runs, proxy-only improves the bootstrap proxy by about

Setting	Method	ΔLoss↓	ΔAcc↑	ΔTail↓
Frozen ERM	SoftClip P95	$+0.055$	$+0.001$	$+\mathbf{5.50}$
	GCE $q = 0.7$	$+0.045$	$+\mathbf{0.004}$	$+\mathbf{5.84}$
	Label Smooth 0.10	—	-0.002	-2.44
	Focal 4	—	-0.004	$-\mathbf{4.11}$
Separate finetune control	SoftClip P95	$+0.221$	$+\mathbf{0.047}$	$+6.63$
	Label Smooth 0.10	-0.276	$+0.029$	$-\mathbf{2.65}$
	Focal 2	$-\mathbf{0.280}$	-0.026	-2.53

Table 3: Within-sweep method-ranking illustration under different reporting criteria. Frozen-ERM rows are relative to the ERM baseline; separate finetune-control rows are relative to that sweep's own baseline finetune checkpoint and are for within-table ranking only. Focal numbers denote γ. On frozen ERM, the positive-accuracy rows are suppressive and tail-worsening; on finetuning, the best-accuracy row is worst on loss and teacher-tail, while label smoothing wins on loss on $10/10$ seeds. Dashes mark held-out loss entries not stored in the original frozen-family static artifact; the shared stored ranking fields for those rows are accuracy and teacher-tail. Bold marks the extreme in each column.

0.010 but worsens held-out hospital-2 loss by $+0.0109$, teacher-tail by $+1.37$, and ECE by $+0.0110$ (Appendix Table 24). A teacher-free held-out stress check gives the same warning: on the top decile of hospital-2 examples fixed by the baseline checkpoint's ordinary cross-entropy loss, proxy-selected SoftClip and GCE increase loss by $+0.94$ and $+0.90$ on $10/10$ seeds (Appendix Table 26). Appendix Table 9 adds a same-stack suppression-strength stress test on the two Camelyon anchors. In the baseline-referenced selected-checkpoint workflow view, Camelyon17 ERM loss / tail harm weakens from P95 to P97 to P99 ($+0.055/+5.50$, $+0.052/+5.16$, $+0.008/+1.76$). Finetuning remains materially harmful across all three selected regimes. Together with the same-trajectory P95/P97 diagnostics, P95 is best read as the designated strong stress-test setting rather than as a knife-edge selector artifact. Appendix Table 19 separately reports the same-trajectory selector-dose diagnostic, where P95/P97 show clean selector-only harm and P99 attenuates. This is supporting rather than core evidence because the fixed $1.25\times$ guardrail is slack on this family, but it shows that the selector mismatch is not unique to one clipping rule.

5.5 The Hazard Concentrates in Suppressive Objectives

As supporting mechanism evidence, Appendix Figure 5 and Table 23 show that tail movement is not uniform across objective families on the same Camelyon17 ERM stack: soft clipping and GCE have $R_w < 1$ and positive tail deltas, while label smoothing and focal loss have $R_w > 1$ and negative tail deltas. On this tested stack, early R_w (epochs 1–5) predicts the sign of the final selected tail delta with $97/100$ seed-level sign accuracy (Appendix Table 27 and Figure 6). The GCE corroboration ($R_w = 0.339$, Appendix Table 25), hard-bootstrap corroboration ($R_w \approx 0.57$, Appendix Table 24), and end-to-end finetune controls (Appendix Table 28) support the same suppress-versus-upweight split. The point is not that any deviation from cross-entropy worsens reliability; it is whether the objective suppresses or amplifies the tail.

The evaluation criterion also changes how objective families rank (Table 3). On frozen Camelyon17 ERM, the positive-accuracy rows are the suppressive SoftClip and GCE checks; GCE $q = 0.7$ has the largest teacher-tail increase ($+5.84$), a 10.0 CVaR swing from Focal 4 (-4.11). On end-to-end finetuning, SoftClip has the highest mean accuracy ($+0.047$) but the worst held-out loss ($+0.221$) and teacher-tail diagnostic ($+6.63$), while label smoothing and focal are lower on loss and tail. Thus accuracy-first and loss/tail-aware readings can favor different methods on the same stack.

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A Appendix

A.1 Mechanistic Formalization

This appendix gives the assumptions and proofs for the three compact propositions stated in Setup. The goal is not a general theorem about deep networks under shift. It is to formalize the selector mismatch created by suppressive weighting, explain why the validation-loss guardrail helps, and give a principled bridge from the empirical time courses to the transient-versus-persistent split.

Proposition 1 (Local lure under suppressive weighting). Assume a suppressive objective whose high-loss slope ratio is $\alpha \in (0, 1)$. Let $A_c > 0$ denote core improvement mass and $A_t > 0$ denote tail damage mass over a short interval. For local loss tradeoffs in which the core improvement is on examples below the clipping threshold and the tail damage is on examples above it, there is a non-empty region in which the suppressive proxy improves while standard risk worsens, namely $\alpha A_t < A_c < A_t$.

Proof. For soft clipping with threshold τ and slope $\alpha \in (0, 1]$,

$$\tilde{\ell} = \ell - (1 - \alpha)(\ell - \tau)_+,$$

so clipping removes upper-tail loss mass directly at the per-example level. When $\tau = \mathrm{VaR}_q(\ell)$, the same monotonicity gives a compressed proxy tail

$$\mathrm{CVaR}_q(\tilde{\ell}) = \tau + \alpha\big(\mathrm{CVaR}_q(\ell) - \tau\big),$$

which is why the proxy can look improved even when the underlying standard-loss tail has not.

Local lure condition. Let $A_c > 0$ denote core improvement mass and $A_t > 0$ denote tail damage mass over a short interval. The short-interval approximation assumes that the core improvement mass lies below the clipping threshold, where the proxy slope is 1, and the tail damage mass lies above it, where the proxy slope is α. Then standard risk and suppressive proxy move as

$$\Delta R \approx -A_c + A_t, \qquad \Delta P \approx -A_c + \alpha A_t.$$

Whenever

$$\alpha A_t < A_c < A_t,$$

the proxy still improves while the standard risk worsens. So the local lure region is non-empty whenever $\alpha < 1$. It widens as α decreases, which is why stronger suppression creates more room for proxy-only selection to move toward less reliable checkpoints. □

Proposition 2 (Baseline-referenced guardrail bound). Assume the guardrail enforces

$$L_{\mathrm{val}}(\hat{\theta}_{\mathrm{guard}}) \le \rho L_{\mathrm{val}}(\hat{\theta}_{\mathrm{base}})$$

and that validation and deployment losses satisfy

$$\sup_t |L_{\mathrm{test}}(\theta_t) - L_{\mathrm{val}}(\theta_t)| \le \varepsilon$$

over the candidate suppressive checkpoints and also for the baseline checkpoint $\hat{\theta}_{\mathrm{base}}$. Then

$$L_{\mathrm{test}}(\hat{\theta}_{\mathrm{guard}}) \le \rho\, L_{\mathrm{test}}(\hat{\theta}_{\mathrm{base}}) + (\rho + 1)\varepsilon.$$

Proof. By the uniform closeness assumption,

$$L_{\mathrm{test}}(\hat{\theta}_{\mathrm{guard}}) \le L_{\mathrm{val}}(\hat{\theta}_{\mathrm{guard}}) + \varepsilon \le \rho\, L_{\mathrm{val}}(\hat{\theta}_{\mathrm{base}}) + \varepsilon.$$

Applying the same bound to the baseline checkpoint gives

$$L_{\mathrm{val}}(\hat{\theta}_{\mathrm{base}}) \le L_{\mathrm{test}}(\hat{\theta}_{\mathrm{base}}) + \varepsilon,$$

so

$$L_{\mathrm{test}}(\hat{\theta}_{\mathrm{guard}}) \le \rho\, L_{\mathrm{test}}(\hat{\theta}_{\mathrm{base}}) + (\rho + 1)\varepsilon.$$

This does not make the guardrail optimal, but it does show why the rule is not arbitrary: the budget caps deployment loss relative to baseline up to validation-to-test mismatch, while proxy-only selection has no analogous guarantee without assuming that the suppressive proxy stays aligned with standard reliability metrics. □

Proposition 3 (Transient versus persistent in a two-phase toy). Let $H_A \geq 0$ denote the excess standard-loss harm accumulated during an early suppressive phase, and let $H_B \geq 0$ denote the later recovery. Then the net hazard is persistent when $H_A > H_B$ and transient when $H_B \geq H_A$.

Proof. By construction, the two-phase toy decomposes the total standard-loss effect into an early excess-harm term and a later recovery term. The sign of the net effect is therefore the sign of $H_A - H_B$. If $H_A > H_B$, the run remains worse than baseline at the matched horizon; if $H_B \geq H_A$, the excess harm has been offset and the matched-horizon hazard disappears. □

Figure 2 illustrates both the local lure region and this two-phase transient-versus-persistent picture.

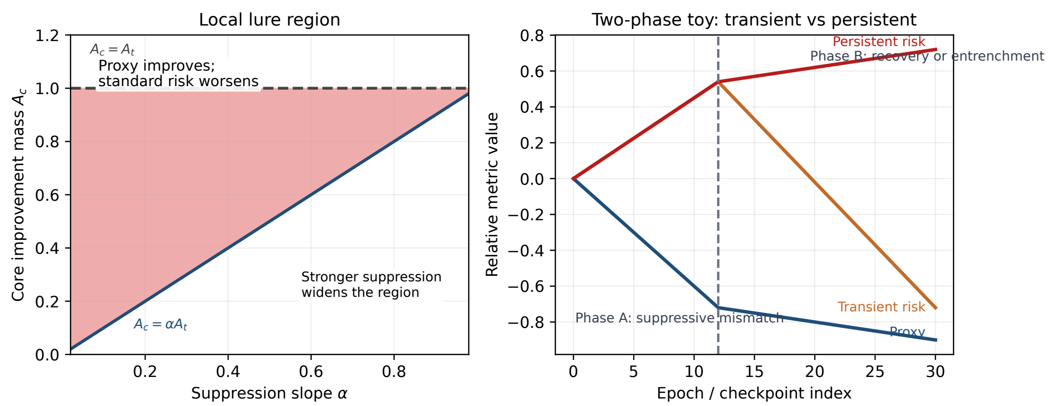

Figure 2: Mechanistic formalization. Left: the local lure region where the suppressive proxy still improves while standard risk worsens. Right: a two-phase toy showing how later recovery yields a transient hazard while entrenched suppression yields a persistent one.

A.2 Persistence Bridge via $R_w(t)$

Table 4 and Figure 3 provide the empirical bridge for the transient-versus-persistent taxonomy. The main discriminator is not a single endpoint value of R_w, but whether suppressive orientation relaxes or becomes more entrenched over training, so we use the early-to-late trend in R_w together with the tail-core clip-gap trend as the primary persistence surrogates.

Suite	Early R_w	Late R_w	ΔR_w	Δclip gap	Trend
Camelyon17	0.277 ± 0.003	0.187 ± 0.002	-0.091 ± 0.003	0.095 ± 0.004	entrenches
CelebA	0.196 ± 0.013	0.298 ± 0.011	0.102 ± 0.017	-0.087 ± 0.013	relaxes

Table 4: Time-course bridge for transient versus persistent hazards. CelebA relaxes over training ($\Delta R_w > 0$ and decreasing tail-core clip gap), while Camelyon17 becomes more entrenched ($\Delta R_w < 0$ and increasing clip gap).

A.3 Core-Case Uncertainty

Table 5 gives paired 95% bootstrap intervals for the three seed-matched hazard anchors used most heavily in the main text. For the Camelyon17 suites, the supporting diagnostic column is teacher-difficulty tail CVaR; for CivilComments, the reliability column is official WILDS worst-group loss.

A.4 Tail-Bank Protocol and Per-Bank Robustness

Because the teacher-tail stress diagnostic is supporting fixed-bank evidence on the vision anchors, we make the bank construction explicit here. The two fixed teacher-difficulty banks are produced once per stack by `src/scripts/build_conf_eval_banks.py`. For Camelyon17, bank A and bank B come from two frozen ERM teacher runs (seed 0 and seed 1, recorded in

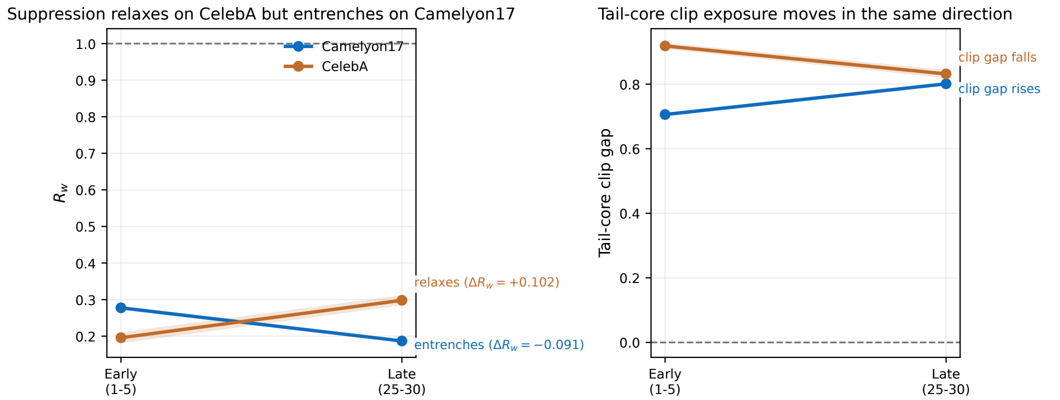

Figure 3: Early-versus-late suppressive orientation on the two main time-course anchors. The trajectories make the transient-versus-persistent split visually explicit: CelebA relaxes, while Camelyon17 entrenches.

Case	ΔHeld-out loss	ΔTail / reliability	ΔECE
Camelyon17 ERM	+0.055 [+0.045, +0.066]	+5.50 [+5.00, +6.04]	+0.032 [+0.030, +0.033]
Camelyon17 Finetune	+0.619 [+0.331, +0.926]	+6.23 [+3.49, +9.07]	+0.021 [+0.018, +0.025]
CivilComments	+0.104 [+0.100, +0.107]	+1.829 [+1.785, +1.871]	+0.070 [+0.068, +0.071]

Table 5: Paired bootstrap intervals for the main hazard anchors. These intervals complement the selected-checkpoint deltas in Table 1 by making the uncertainty on held-out loss, reliability, and calibration explicit.

`artifacts/partitions_eval/camelyon17_resnet50/teacher_difficulty/meta.json`). Within each bank, examples are assigned to $K = 64$ quantile cells by per-example teacher cross-entropy loss, with a minimum evaluation cell size of 20 enforced downstream by `src/scripts/phase1_eval.py`. Those bank assignments are fixed before selector comparison and are not redefined per checkpoint. Table 6 and Table 7 then rerun the selected-checkpoint comparison separately on banks A and B using `src/scripts/analyze_per_bank_robustness.py`. For Camelyon17 ERM, the harmful direction holds cleanly on both banks at P95, P97, and P99. For finetuning, the selected-checkpoint bankwise direction remains at P95, P97, and P99 as well: the harmful mean tail direction holds on both banks, with slightly weaker decoupling fractions at P95 on both banks and P97 on bank B.

Dataset	Bank	Regime	ΔProxy	ΔTail	Proxy improve frac	Decoupling frac
Camelyon17	A	P95	-0.297	+5.501	1.00	1.00
Camelyon17	A	P97	-0.267	+5.155	1.00	1.00
Camelyon17	A	P99	-0.151	+1.765	1.00	1.00
Camelyon17	B	P95	-0.258	+5.501	1.00	1.00
Camelyon17	B	P97	-0.170	+5.155	1.00	1.00
Camelyon17	B	P99	-0.029	+1.765	1.00	1.00

Table 6: Camelyon17 ERM per-bank robustness for the teacher-difficulty tail metric. Rows report mean proxy and tail deltas relative to the ERM baseline separately on banks A and B. The harmful teacher-difficulty tail direction holds on both banks, so the main P95 result is not driven by averaging across a single favorable bank. For ERM, the bankwise tail deltas coincide after rounding in the saved artifact, while proxy deltas differ by bank.

Dataset	Bank	Regime	ΔProxy	ΔTail	Proxy improve frac	Decoupling frac
Camelyon17 finetune selected	A	P95	-0.605	+6.430	1.00	0.90
Camelyon17 finetune selected	A	P97	-0.528	+5.409	1.00	1.00
Camelyon17 finetune selected	A	P99	-0.359	+5.744	1.00	1.00
Camelyon17 finetune selected	B	P95	-0.458	+6.022	1.00	0.90
Camelyon17 finetune selected	B	P97	-0.391	+5.484	1.00	0.90
Camelyon17 finetune selected	B	P99	-0.261	+5.653	1.00	1.00

Table 7: Camelyon17 finetune selected-checkpoint per-bank robustness for the teacher-difficulty tail metric. The harmful tail direction remains on both banks at P95, P97, and P99. P95 and P97 are slightly less uniform than the ERM anchor on one of the two bankwise consistency fractions, while P99 remains fully harmful on both banks.

A.5 Persistent-Case Fixed-Horizon Support

Table 8 exposes the matched fixed-horizon comparison for every case we still label persistent in the main paper. For the classification anchors, the matched horizons are epoch 30 (Camelyon17 ERM and frozen CivilComments) and epoch 10 (Camelyon17 finetuning), matching the main-text setup. For ACSIncome, all runs share the same 50-epoch training horizon, so the fixed-horizon comparison uses the final epoch. The goal is not to claim identical magnitudes between selected and fixed views; it is to show that the harmful loss / reliability direction survives the matched-horizon check for every row that retains the persistent label. For Camelyon17 finetuning, the fixed10 teacher-difficulty tail diagnostic is markedly more outlier-sensitive than the selected-checkpoint view, so we use it only as a qualitative persistence check rather than as a clean magnitude claim.

Case	Fixed horizon	n	ΔLoss (sel.)	ΔLoss (fixed)	ΔRel. (sel.)	ΔRel. (fixed)	Verdict
Camelyon17 ERM	ep30	10	+0.055	+0.053	+5.501	+5.434	Persistent
Camelyon17 Finetune	ep10	10	+0.619	+0.491	+6.226	mean $> 0^{\dagger}$	Persistent
Frozen CivilComments	ep30	10	+0.104	+0.124	+1.829	+1.923	Persistent
ACSIncome	ep50	10	+0.027	+0.017	+0.376	+0.284	Persistent

Table 8: Fixed-horizon support for every row still labeled persistent. "Selected" is the main-text proxy-selected comparison; "fixed" evaluates the same baseline/suppressive pair at the matched horizon instead of at the selected epoch. For ACSIncome, the matched fixed horizon is the common final epoch 50. † The finetune fixed10 teacher-difficulty tail diagnostic is outlier-sensitive: a few seeds explode at epoch 10, so we use that entry only as a qualitative persistence check rather than as a clean magnitude comparison.

A.6 Suppression-Strength Dose Response on Camelyon

Table 9 adds a direct suppression-strength stress test on the two strongest Camelyon anchors. For ERM, weakening the clipping threshold from P95 to P97 to P99 reduces the baseline-referenced selected-checkpoint harm smoothly in both held-out loss and teacher-difficulty tail CVaR, so the main P95 result is not a knife-edge artifact of one particular threshold. For finetuning, the selected-checkpoint harm remains large across all three regimes, and the bankwise selected-checkpoint comparison continues to show harmful tail movement on both banks (Table 7). The matched fixed10 finetune view is materially noisier, so we do not present the finetune dose-response as a monotone law; the supportable conclusion is narrower: once suppressive finetuning is allowed to select by its own proxy, the selected-checkpoint hazard persists beyond a single P95 setting.

A.7 Guardrail Uncertainty

Table 10 reports paired bootstrap intervals for the improvement achieved by the guardrail relative to proxy-only selection. The intervals are computed on seed-matched proxy-only versus guardrail selections.

Suite	Regime	Frac clip	ΔLoss↓	ΔTail↓
Camelyon17 ERM	P95	0.089	+0.055	+5.50
	P97	0.076	+0.052	+5.16
	P99	0.037	+0.008	+1.76
Camelyon17 finetune	P95	0.068	+0.619	+6.23
	P97	0.060	+0.772	+5.45
	P99	0.043	+0.551	+5.70

Table 9: Camelyon suppression-strength dose response on the selected checkpoints. Frac clip reports the mean validation clipping fraction at the selected epoch. In the baseline-referenced selected-checkpoint workflow view, ERM loss / tail harm weakens as clipping relaxes from P95 to P97 to P99. Table 19 separately reports the same-trajectory selector-dose diagnostic. On finetuning, the selected-checkpoint hazard remains large across all three tested clipping thresholds, so P95 should be read as the designated strong stress-test setting rather than as a unique knife-edge.

Suite	Budget	Loss reduction	Tail reduction	Acc. change
Camelyon17 ERM	1.10x	+0.055 [+0.046,+0.066]	+5.501 [+5.001,+6.022]	-0.001 [-0.002,+0.001]
Camelyon17 ERM	1.25x	+0.029 [+0.016,+0.044]	+2.988 [+1.873,+4.209]	-0.003 [-0.004,-0.001]
Camelyon17 ERM	1.50x	+0.003 [+0.000,+0.010]	+0.177 [+0.000,+0.532]	+0.000 [+0.000,+0.001]
Camelyon17 Finetune	1.10x	+0.626 [+0.310,+0.956]	+6.382 [+3.712,+9.171]	-0.024 [-0.074,+0.027]
Camelyon17 Finetune	1.25x	+0.551 [+0.260,+0.873]	+5.767 [+2.684,+8.879]	-0.015 [-0.057,+0.027]
Camelyon17 Finetune	1.50x	+0.301 [+0.061,+0.580]	+4.234 [+1.036,+7.813]	-0.025 [-0.058,+0.006]

Table 10: Paired bootstrap intervals for proxy-only → guardrail changes. Positive loss/tail reductions mean the guardrail removes harm relative to proxy-only selection.

A.8 Guardrail Budget Sweep Detail

Table 11 retains the full budget sweep that sits behind the main-text selector comparison. It shows why the main paper centers the selector comparison on the $1.25\times$ operating point as a middle diagnostic budget rather than presenting it as a universal constant.

Suite	Rule	Accept rate	ΔHeld-out loss↓	ΔHeld-out acc↑	ΔTail CVaR↓
Camelyon17 ERM	Proxy-only	1.00	+0.055	+0.001	+5.501
	1.10x budget	0.00	+0.000	+0.000	+0.000
	1.25x budget	0.70	+0.026	-0.002	+2.513
	1.50x budget	1.00	+0.052	+0.001	+5.324
Camelyon17 Finetune	Proxy-only	1.00	+0.619	+0.025	+6.226
	1.10x budget	0.10	+0.000	+0.000	-0.156
	1.25x budget	0.40	+0.068	+0.010	+0.460
	1.50x budget	0.70	+0.318	+0.000	+1.992

Table 11: Full budget sweep on the two Camelyon17 anchor suites. The $1.10\times$ budget is essentially a veto, $1.50\times$ is materially looser, and $1.25\times$ is the middle illustrative diagnostic budget used in the main text.

A.9 Adaptive Baseline-Only Guardrail Sensitivity

To address reviewer-facing concerns about ρ, we also checked a fully validation-only adaptive heuristic. Instead of using the fixed $1.25\times$ budget from Equation 3, it sets the admissible threshold to $\mu + 2\sigma$ from the final stable baseline-loss window and then applies proxy-based selection only within that feasible set. Table 12 shows that the result is conservative rather than contradictory: on Camelyon17 ERM it becomes a pure veto, while on finetuning it narrows but does not eliminate proxy-only damage. We therefore treat adaptive baseline-only budgets as a sensitivity analysis, not as a replacement for the illustrative fixed-budget operating point used in the main selector comparison.

Suite	Selector	Admit rate	ΔHeld-out loss↓	ΔHeld-out acc↑	ΔTail↓
Camelyon17 ERM	Baseline	1.00	+0.000	+0.000	+0.000
	Proxy-only	1.00	+0.055	+0.001	+5.501
	Adaptive $\mu + 2\sigma$	0.00	+0.000	+0.000	+0.000
Camelyon17 Finetune	Baseline	1.00	+0.000	+0.000	+0.000
	Proxy-only	1.00	+0.619	+0.025	+6.226
	Adaptive $\mu + 2\sigma$	0.90	+0.482	+0.031	+5.190

Table 12: Baseline-only adaptive guardrail sensitivity on the Camelyon17 selector anchors. The adaptive threshold is set from final baseline validation-loss fluctuations alone ($\mu + 2\sigma$; last five epochs on 30-epoch ERM and last three epochs on 10-epoch finetuning). Deltas are measured relative to the baseline-selected checkpoint. The heuristic is conservative on ERM and still directionally safer than proxy-only on finetuning, but it is not as effective as the illustrative fixed $1.25\times$ operating point from the main text.

A.10 Public-Practice Audit of Scalarized or Under-Specified Selection

We audited 10 adjacent published papers with released artifacts, spanning benchmark/selection artifacts and shaped-objective or robust-learning repos (DomainBed, WILDS, SWAD, SubpopBench, ERM++, group DRO [31], To Smooth or Not? [35], LS-PLL [14], Symmetric Cross Entropy [34], and SGN [10]). The sample is illustrative only, chosen to span adjacent released benchmark/selection artifacts and suppressive-or-robust objective repos rather than to estimate frequency or representativeness. Table 13 shows the pattern: $8/10$ expose checkpoint choice through a single best-model scalar, and none exposes a default reliability-aware multi-metric selector or checks alignment with held-out loss, calibration, or group-sensitive reliability.

Artifact	Family	Exposed best-model rule	Scalarized default?	Loss/calib cross-check?
DomainBed	DG	top `val_acc`	Yes	No
WILDS runner	shift	best epoch by `val_metric`	Yes	No
group_DRO	robust rwgt.	worst-group / avg. val acc	Yes	No
SWAD	DG	in-domain val acc	Yes	No
SubpopBench	subgroup	val worst-group / worst-class acc	Yes	No
ERM++	DG	highest val acc	Yes	No
To Smooth or Not?	label smooth.	held-out acc	Yes	No
LS-PLL	partial labels	test acc	Yes	No
Symmetric CE	noisy labels	epochwise `val_loss`	Yes	No
SGN	noisy labels	no default best-model rule	No	No

Table 13: Compact audit of checkpoint-selection defaults in adjacent published papers with released artifacts. The sample spans benchmark/selection artifacts and shaped-objective or robust-learning repos. This is a purposive relevance audit rather than a prevalence estimate; the $8/10$ count should not be read as a population rate. The common pattern is not a specific metric, but that checkpoint choice is scalarized or left under-specified by default rather than exposed through a reliability-aware multi-metric selector.

A.11 End-to-End CivilComments Support

As a targeted scope check, we also reran CivilComments end-to-end with DistilBERT finetuning on an official-group-aware skewed validation surface emphasizing minority-identity positive examples. Table 14 reports the ten-seed selector comparison. Relative to the baseline-selected checkpoint, proxy-only soft clipping worsens overall loss by $+0.155$, official WILDS worst-group loss by $+1.007$, and ECE by $+0.0533$; validation-loss-only remains worse than baseline, while the $1.25\times$ guardrail acts as a pure veto on all $10/10$ seeds.

Unlike the frozen-embedding CivilComments result, fixed-horizon official worst-group persistence is mixed in this end-to-end text stack: at epoch 10, soft clipping is still worse on overall loss and ECE, but only $5/10$ seeds are worse on official worst-group loss and worst-group accuracy is often higher $(8/10)$. So this extension stays in the appendix; its value is narrower, namely that the selected-checkpoint loss / calibration hazard survives end-to-end text finetuning even when the fixed-horizon worst-group signal is mixed, consistent with early suppressive R_w relaxing toward 1 later in training (Figure 4).

Selector	ΔAcc	ΔLoss	ΔWG Acc	ΔWG Loss	ΔECE
Proxy-only	-0.0046	+0.155	+0.0330	+1.007	+0.0533
Val-loss-only	-0.0022	+0.096	-0.0583	+0.919	+0.0471
Guardrail $1.25\times$	+0.0000	+0.000	+0.0000	+0.000	+0.0000

Table 14: End-to-end CivilComments selector comparison on the official-group-aware validation surface. Deltas are measured relative to the baseline-selected checkpoint. Proxy-only soft clipping worsens overall loss and official WILDS worst-group loss, while the $1.25\times$ guardrail acts as a pure veto and falls back to baseline on all ten seeds. Rows are backed by the saved selector-summary artifact listed in the supplement ledger.

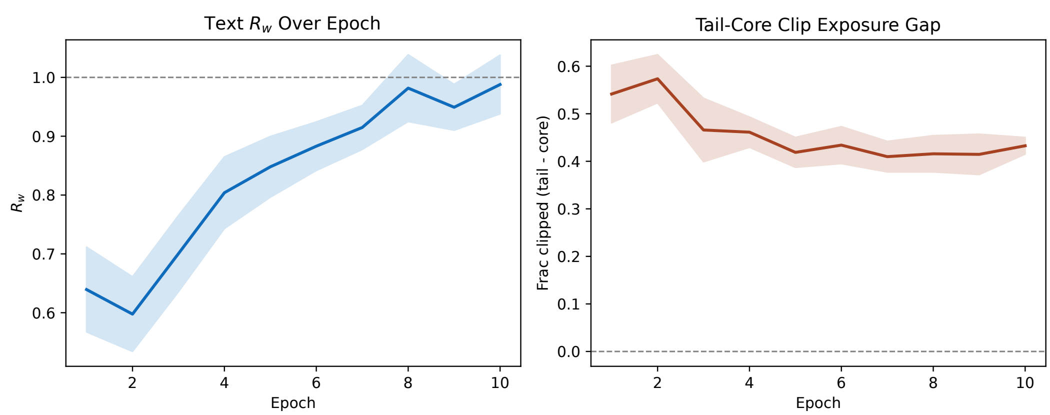

Figure 4: End-to-end CivilComments mechanism on the official-group-aware validation surface. Left: early R_w is suppressive before relaxing toward 1 later in training. Right: the tail-core clip-exposure gap remains large, showing that the selected checkpoints still sit in a genuinely suppressive regime.

A.12 ACSIncome Huber Dose-Response

On the same ACSIncome regression stack, Huber loss at the P95 residual threshold provides a milder suppressive objective ($R_w = 0.53$, vs. 0.41 for winsorization). The selection hazard is weaker and shifts toward the tail channel: proxy-only selection worsens raw held-out MSE on $3/10$ seeds and tail MSE on $9/10$ seeds. The mean Δ tail MSE is $+0.051$ with 95% CI $[+0.028, +0.073]$ (excludes zero), while the mean Δ held-out MSE is slightly negative at -0.0031 with 95% CI $[-0.0053, -0.0008]$. This is consistent with Proposition 1's dose-response picture: Huber preserves a weaker but non-zero tail signal, and the ordering $R_w^{\text{wins}} = 0.41 < R_w^{\text{huber}} = 0.53$ maps to a much stronger winsorization hazard.

A.13 Seed-Matched Lure Checks

The main paper reports the strongest seed-level counts in prose. Table 15 gives the full dominance counts, while Table 16 gives paired mean deltas with bootstrap intervals. For Camelyon17, the reliability diagnostic is teacher-difficulty tail CVaR; for CivilComments, it is official WILDS worst-group loss.

Suite	Proxy better	Proxy∧val loss worse	Proxy∧test loss worse	Proxy∧reliability worse	Proxy∧ECE worse
Camelyon17 ERM	**10/10**	**10/10**	**10/10**	**10/10**	**10/10**
Camelyon17 Finetune	8/10	8/10	7/10	7/10	8/10
CivilComments	**10/10**	**10/10**	**10/10**	**10/10**	**10/10**

Table 15: Seed-level lure counts. "Reliability worse" means higher teacher-difficulty tail CVaR on Camelyon17 and higher official WILDS worst-group loss on CivilComments.

Suite	Metric	Mean	95% CI low	95% CI high
Camelyon17 ERM	proxy clip	-0.0709	-0.0769	-0.0659
Camelyon17 ERM	validation loss	+0.0891	+0.0762	+0.1034
Camelyon17 ERM	held-out loss	+0.0553	+0.0455	+0.0658
Camelyon17 ERM	tail CVaR	+5.5015	+5.0025	+6.0403
Camelyon17 ERM	ECE	+0.0321	+0.0304	+0.0334
Camelyon17 ERM	Brier	+0.0027	+0.0014	+0.0038
Camelyon17 ERM	held-out acc	+0.0010	-0.0007	+0.0025
Camelyon17 Finetune	proxy clip	-0.0141	-0.0377	+0.0119
Camelyon17 Finetune	validation loss	+0.1229	+0.0754	+0.1733
Camelyon17 Finetune	held-out loss	+0.6185	+0.3311	+0.9258
Camelyon17 Finetune	tail CVaR	+6.2264	+3.4939	+9.0689
Camelyon17 Finetune	ECE	+0.0214	+0.0178	+0.0249
Camelyon17 Finetune	Brier	+0.0006	-0.0033	+0.0051
Camelyon17 Finetune	held-out acc	+0.0251	-0.0218	+0.0700
CivilComments	proxy clip	-0.0143	-0.0151	-0.0135
CivilComments	validation loss	+0.0609	+0.0583	+0.0638
CivilComments	held-out loss	+0.1037	+0.1002	+0.1071
CivilComments	ECE	+0.0701	+0.0685	+0.0714
CivilComments	Brier	+0.0138	+0.0134	+0.0143
CivilComments	held-out acc	-0.0040	-0.0045	-0.0034
CivilComments	WILDS wg loss	+1.8292	+1.7853	+1.8710
CivilComments	WILDS wg acc	-0.1474	-0.1545	-0.1394
CivilComments	WILDS wg Brier	+0.3003	+0.2945	+0.3061
CivilComments	WILDS wg ECE	+0.3041	+0.2952	+0.3126

Table 16: Paired mean deltas for the main lure checks. "Reliability" denotes teacher-difficulty tail CVaR on Camelyon17 and official WILDS worst-group loss on CivilComments; calibration is reported through both ECE and Brier.

A.14 Camelyon Standard-Metric, Selector-Dose, and Leave-One-Hospital Diagnostics

Table 17 adds two targeted diagnostics for the Camelyon anchors. The workflow rows show that the Camelyon reliability signal is not carried only by teacher-difficulty tail CVaR: on ERM, held-out loss, ECE, and Brier all worsen; on finetuning, held-out loss and validation-side ECE worsen while Brier is neutral. The same-trajectory rows compare proxy-best with validation-loss-best inside the same suppressive trajectory, with no fallback. This SoftClip P95 selector-only contrast is clean on Camelyon17 ERM and mixed on the main finetuning anchor, so the SoftClip finetuning headline remains primarily baseline-referenced workflow evidence. Table 18 reports a separate end-to-end

GCE ($q = 0.7$) finetune selector audit with clean held-out-loss and fixed loss-tail degradation relative to validation-loss-best.

Suite	Contrast	ΔLoss	ΔAcc	ΔECE	ΔBrier	ΔTail	Harm counts
Camelyon17 ERM	workflow: proxy - baseline	+0.055 [+0.046,+0.066]	+0.001 [-0.001,+0.003]	+0.0321 [+0.0305,+0.0334]	+0.0027 [+0.0013,+0.0038]	+5.50 [+5.00,+6.04]	loss/ECE/tail 10/10,10/10,10/10; Brier 8/10
Camelyon17 Finetune	workflow: proxy - baseline	+0.619 [+0.328,+0.916]	+0.025 [-0.021,+0.071]	+0.0214 [+0.0177,+0.0248]	+0.0006 [-0.0034,+0.0051]	+6.23 [+3.46,+9.06]	loss/ECE/tail 9/10,10/10,9/10; Brier 4/10
Camelyon17 ERM	same trajectory: proxy - val-loss	+0.020 [+0.011,+0.030]	+0.003 [+0.002,+0.005]	+0.0068 [+0.0047,+0.0090]	-0.0011 [-0.0023,+0.0001]	+2.01 [+1.47,+2.60]	loss/ECE/tail 10/10,10/10,10/10; Brier 3/10
Camelyon17 Finetune	same trajectory: proxy - val-loss	+0.010 [-0.135,+0.145]	+0.031 [+0.008,+0.055]	-0.0026 [-0.0081,+0.0029]	-0.0050 [-0.0118,+0.0008]	+3.42 [+0.75,+6.13]	loss/ECE/tail 2/10,1/10,4/10; Brier 1/10

Table 17: Camelyon standard-metric triangulation and same-trajectory selector-only contrasts. Workflow rows compare the proxy-selected suppressive checkpoint with the baseline-selected undistorted checkpoint. Same-trajectory rows compare proxy-best with validation-loss-best on the same suppressive P95 trajectory and do not use fallback. ΔTail is teacher-difficulty tail CVaR, used here as a fixed-bank stress diagnostic. For Camelyon17 finetune rows, ECE/Brier are validation-side calibration diagnostics from the saved finetune selector-calibration artifact; the ERM rows use held-out/test calibration. Workflow-row intervals are regenerated in this standard-metric triangulation artifact; small endpoint differences from Table 5 reflect separate bootstrap resampling of the same seed-matched deltas.

Table 18 adds a narrow end-to-end same-trajectory selector audit using GCE ($q = 0.7$) on Camelyon17 finetuning. The comparison is proxy-best minus validation-loss-best on the same ten GCE finetune trajectories. It supports the loss/tail selector warning beyond the frozen-ERM stack: proxy-best improves the objective-aligned validation proxy and validation accuracy, but worsens held-out hospital-2 loss and a teacher-free held-out loss-tail. Held-out accuracy, ECE, and Brier are mixed, so we do not use this table as a held-out accuracy or calibration claim.

Metric	Δ proxy-best − val-loss-best	Count	Readout
GCE validation proxy	-0.0153 [-0.0190,-0.0115]	10/10 proxy better	proxy improves
Validation CE loss	+0.148 [+0.113,+0.184]	10/10 worse	validation loss worsens
Validation accuracy	+0.0046 [+0.0007,+0.0086]	8/10 better	validation accuracy improves
Held-out loss	+0.566 [+0.157,+0.974]	8/10 worse	held-out loss worsens
Fixed held-out loss-tail	+3.49 [+1.69,+5.28]	10/10 worse	tail worsens
Held-out accuracy	+0.023 [-0.021,+0.068]	4/10 better	mixed
Held-out ECE	+0.014 [-0.035,+0.064]	6/10 worse	mixed
Held-out Brier	+0.002 [-0.040,+0.044]	6/10 worse	mixed

Table 18: End-to-end GCE ($q = 0.7$) same-trajectory selector audit on Camelyon17 finetuning. Deltas are proxy-best minus validation-loss-best on the same ten GCE finetune trajectories. Negative proxy means the objective-aligned validation score improves. Proxy-best improves the GCE validation proxy and validation accuracy, but worsens ordinary validation CE, held-out hospital-2 loss, and a fixed teacher-free held-out loss-tail. Held-out accuracy, ECE, and Brier are mixed, so this table is used only as loss/tail same-trajectory selector support, not as a held-out-accuracy or calibration claim.

Table 19 extends the Camelyon17 ERM same-trajectory selector diagnostic beyond the P95 stress setting. Holding the suppressive objective and trajectory fixed, proxy-best selection is worse than validation-loss-best selection at P95 and P97 on held-out loss, ECE, and the fixed-bank teacher-tail diagnostic. At P99 the result attenuates and becomes mixed: standard loss and Brier no longer worsen, but ECE and teacher-tail still do. This is why we treat P95/P97 as clean selector-only evidence and P99 as an attenuation check rather than as a contradiction.

Regime	ΔLoss	ΔECE	ΔBrier	ΔTail	Harm counts	Note
P95	+0.020 [+0.011,+0.030]	+0.0068 [+0.0047,+0.0091]	-0.0011 [-0.0024,+0.0002]	+2.01 [+1.46,+2.61]	loss/ECE/tail 10/10,10/10,10/10; Brier 3/10	clean
P97	+0.023 [+0.018,+0.028]	+0.0105 [+0.0088,+0.0122]	-0.0013 [-0.0025,-0.0002]	+2.39 [+2.05,+2.71]	loss/ECE/tail 10/10,10/10,10/10; Brier 2/10	clean
P99	-0.003 [-0.006,-0.001]	+0.0050 [+0.0029,+0.0071]	-0.0023 [-0.0035,-0.0012]	+0.71 [+0.42,+1.02]	loss/ECE/tail 1/10,8/10,8/10; Brier 0/10	attenuated/mixed

Table 19: Camelyon17 ERM same-trajectory selector-dose diagnostic. Deltas are proxy-best minus validation-loss-best on the same suppressive trajectory, so this isolates selector choice within a fixed objective/trajectory rather than a baseline-referenced workflow comparison. P95 and P97 show clean selector-only harm on held-out loss, ECE, and the fixed-bank teacher-tail diagnostic. P99 is attenuated and mixed: loss and Brier no longer worsen, but ECE and teacher-tail still degrade.

Table 20 checks validation-accuracy selection, a common scalarized checkpoint rule. The comparison is validation-accuracy-best minus validation-loss-best on the same Camelyon17 ERM suppressive trajectory. It is not a new workflow-generalization claim; it shows that the same practical selector warning is not limited to the soft-clipped proxy name.

Regime	ΔLoss	ΔECE	ΔBrier	ΔTail	Harm counts	Note
P95	+0.023 [+0.017,+0.030]	+0.0072 [+0.0060,+0.0084]	-0.0020 [-0.0031,-0.0009]	+2.39 [+2.08,+2.72]	loss/ECE/tail 10/10,10/10,10/10; Brier 2/10	clean loss/ECE/tail
P97	+0.027 [+0.023,+0.030]	+0.0117 [+0.0103,+0.0132]	-0.0027 [-0.0037,-0.0018]	+2.92 [+2.70,+3.19]	loss/ECE/tail 10/10,10/10,10/10; Brier 0/10	clean loss/ECE/tail
P99	+0.026 [+0.019,+0.031]	+0.0221 [+0.0178,+0.0256]	+0.0002 [-0.0008,+0.0012]	+2.90 [+2.36,+3.33]	loss/ECE/tail 10/10,10/10,10/10; Brier 5/10	clean loss/ECE/tail

Table 20: Camelyon17 ERM validation-accuracy selector sensitivity. Deltas are validation-accuracy-best minus validation-loss-best on the same suppressive trajectory. P95, P97, and P99 all worsen held-out loss, ECE, and the fixed-bank teacher-tail diagnostic in $10/10$ seeds. Brier is neutral or favorable to validation-accuracy selection, so the claim is restricted to loss, calibration by ECE, and the teacher-tail stress diagnostic.

Table 21 adds a deliberately narrow same-trajectory selector audit and checkpoint-averaging sensitivity, not a benchmark of SWAD or other averaging protocols. Proxy-best, validation-accuracy selection, and a last-5 checkpoint average remain worse than validation-loss selection on held-out loss, ECE, and the teacher-tail diagnostic for both SoftClip P95 and GCE $q = 0.7$. A validation-loss-neighborhood average mitigates held-out loss and ECE and leaves tail movement neutral, so averaging is best read here as a mitigation around the validation-loss rule rather than evidence that proxy-selected reporting is reliable.

Objective	Policy − val-loss-best	ΔLoss	ΔECE	ΔTail	Harm counts
SoftClip P95	Proxy-best	+0.020 [+0.011,+0.030]	+0.0068 [+0.0047,+0.0090]	+2.01 [+1.46,+2.59]	10/10,10/10,10/10
	Val-acc-best	+0.023 [+0.017,+0.030]	+0.0072 [+0.0059,+0.0084]	+2.39 [+2.08,+2.72]	10/10,10/10,10/10
	Last-5 avg.	+0.037 [+0.036,+0.038]	+0.0101 [+0.0094,+0.0108]	+3.24 [+3.01,+3.48]	10/10,10/10,10/10
	Val-loss-neigh. avg.	-0.001 [-0.003,-0.000]	-0.0007 [-0.0011,-0.0003]	-0.01 [-0.08,+0.08]	2/10,1/10,5/10
GCE q=0.7	Proxy-best	+0.018 [+0.014,+0.023]	+0.0129 [+0.0105,+0.0154]	+3.05 [+2.51,+3.59]	10/10,10/10,10/10
	Val-acc-best	+0.031 [+0.021,+0.041]	+0.0156 [+0.0129,+0.0182]	+3.77 [+3.22,+4.29]	10/10,10/10,10/10
	Last-5 avg.	+0.053 [+0.052,+0.055]	+0.0204 [+0.0187,+0.0223]	+4.96 [+4.68,+5.22]	10/10,10/10,10/10
	Val-loss-neigh. avg.	-0.005 [-0.006,-0.003]	-0.0017 [-0.0028,-0.0003]	+0.05 [-0.06,+0.22]	1/10,2/10,4/10

Table 21: Same-trajectory selector audit and checkpoint-averaging sensitivity on Camelyon17 ERM. Rows report policy minus validation-loss-best on the same suppressive objective family and trajectory. For both SoftClip P95 and GCE ($q = 0.7$), proxy-best, validation-accuracy-best, and last-5 averaging worsen held-out loss, ECE, and the teacher-tail stress diagnostic in $10/10$ seeds, while validation-loss-neighborhood averaging mitigates loss/ECE and leaves tail movement approximately neutral. This is a narrow sensitivity on released runs, not a SWAD or Ensemble-of-Averages benchmark.

Table 22 repeats a standard-metric selector-only check over leave-one-hospital-out Camelyon17 ERM folds. Because the comparison is proxy-best minus validation-loss-best on the same suppressive trajectory, it supports selector-only robustness. We do not use it to claim that proxy-minus-baseline workflow harm is uniform across hospitals.

Regime	ΔLoss	ΔAcc	ΔBrier	ΔECE	Harmful folds
P95	+0.020	-0.0030	+0.0028	+0.0057	loss/Brier 5/5,5/5; acc/ECE 5/5,5/5
P97	+0.011	-0.0014	+0.0018	+0.0040	loss/Brier 5/5,5/5; acc/ECE 4/5,4/5
P99	+0.008	-0.0014	+0.0019	+0.0067	loss/Brier 5/5,5/5; acc/ECE 5/5,5/5

Table 22: Leave-one-hospital-out Camelyon17 ERM selector-only standard-metric check. Deltas are proxy-best minus validation-loss-best on the same suppressive trajectory, averaged over five held-out hospital folds. These rows use standard held-out metrics only and do not use the teacher-tail diagnostic. They are included as a selector-only robustness check, not as evidence that the baseline-referenced proxy-minus-baseline workflow hazard generalizes uniformly across hospitals.

A.15 Objective-Family Numerical Controls

Figure 5 and Table 23 give the mechanism view and numerical values behind the objective-family discussion.

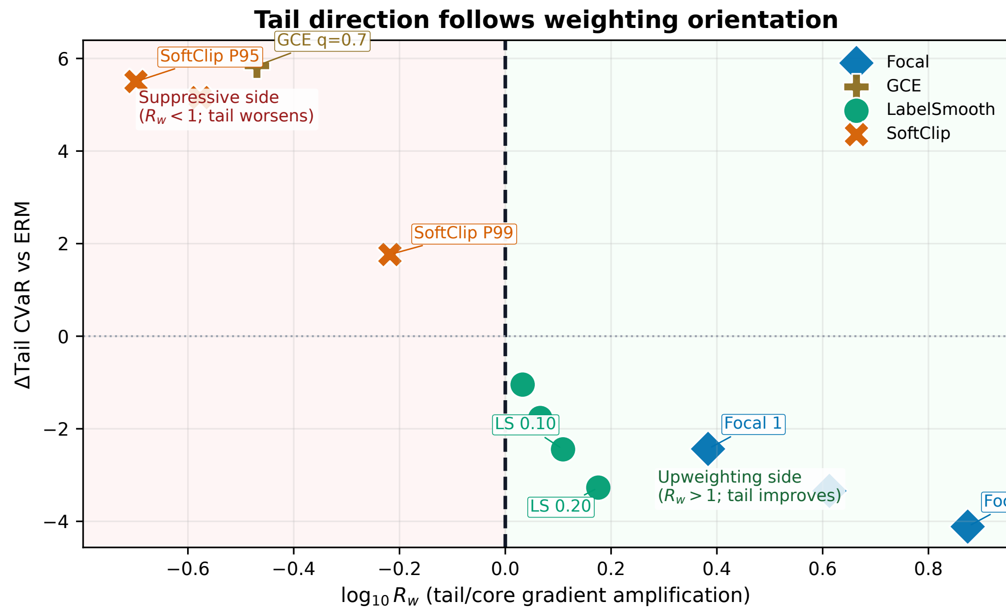

Figure 5: Same-stack objective-family mechanism on Camelyon17 ERM. The vertical boundary at $\log_{10} R_w = 0$ separates tail suppression ($R_w < 1$) from tail upweighting ($R_w > 1$). Focal numbers denote γ; soft clipping and GCE lie on the suppressive side and inflate the tail, while focal loss and label smoothing lie on the upweighting side and improve it.

A.16 Hard-Bootstrap Corroboration

Table 24 gives the fuller second-family selector study on the same Camelyon17 ERM stack using hard bootstrapped cross-entropy with $\beta = 0.60$, a standard suppressive objective from the noisy-label literature. The core selector mismatch survives this extension. Relative to val-loss-only selection inside the same bootstrap runs, proxy-only improves the bootstrap proxy by about 0.010 but worsens held-out hospital-2 loss by $+0.0109$, teacher-difficulty tail CVaR by $+1.37$, and ECE by $+0.0110$ on average across ten seeds. The selected checkpoints remain suppressive ($R_w \approx 0.567$, early $R_w \approx 0.605$).

The fixed $1.25\times$ guardrail is effectively slack on this family, while the within-run oracle-loss selector lands much earlier (mean epoch 6.4 versus 15.2 for proxy-only) and is lower on held-out loss, teacher-difficulty tail CVaR, and ECE. Unlike the main P95 soft-clipping anchor, this bootstrap corroboration is not a stronger fixed-horizon pathology: relative to fixed epoch 30, the selected bootstrap checkpoint is actually better on proxy, held-out loss, teacher-difficulty tail CVaR, and ECE. So we use it as supporting evidence against the "only soft clipping" objection rather than as a replacement for the main hazard anchors.

A.17 Generalized Cross-Entropy Corroboration

Table 25 gives a second standard noisy-label objective check on the Camelyon17 ERM stack. Generalized cross-entropy with $q = 0.7$ is suppressive on this stack, and the objective-aligned proxy selector moves to later checkpoints than validation-loss selection. The result is supporting evidence, not a new optimizer comparison: it shows that the soft-clipping hazard also appears for a standard suppressive objective family. We use $q = 0.7$ as a single mid-range corroborating setting rather than a tuned GCE sweep.

A.18 Teacher-Free Held-Out Loss-Tail Check

Table 26 gives a teacher-free stress check for the two strongest suppressive Camelyon17 ERM cases. For each seed and objective, we fix the top 10% of held-out hospital-2 examples by the baseline-

Setting	ΔProxy↓	ΔTail CVaR↓	ΔHeld-out acc↑	R_w
Focal 1	-0.055	-2.435	-0.0016	2.417
Focal 2	-0.066	-3.342	-0.0018	4.100
Focal 4	-0.070	-4.113	-0.0035	7.485
GCE q=0.7	-0.024	+5.843	+0.0039	0.339
LS 0.02	-0.025	-1.046	-0.0005	1.078
LS 0.05	-0.029	-1.768	-0.0016	1.164
LS 0.10	-0.033	-2.444	-0.0022	1.286
LS 0.20	-0.040	-3.271	-0.0019	1.499
SoftClip P95	-0.224	+5.501	+0.0010	0.200
SoftClip P97	-0.098	+5.155	-0.0009	0.265
SoftClip P99	+0.063	+1.765	-0.0009	0.605

Table 23: Same-stack objective-family controls on Camelyon17 ERM. Soft clipping and GCE suppress the tail ($R_w < 1$), while focal loss and label smoothing upweight it ($R_w > 1$). Focal 1/2/4 denote focal-loss focusing parameter $\gamma = 1, 2, 4$. The R_w values here are selected-checkpoint orientation values; Appendix Table 27 and Figure 6 separately report early-window R_w for the warning-statistic analysis.

Selector	ΔHeld-out loss↓	ΔHeld-out acc↑	ΔTail CVaR↓	ΔECE↓
Baseline	+0.000	+0.000	+0.000	+0.000
Proxy-only	+0.016	-0.003	+1.892	+0.008
Val-loss-only	+0.005	-0.002	+0.519	-0.003
1.25x guardrail	+0.016	-0.003	+1.892	+0.008
Oracle loss	-0.000	+0.003	+0.921	+0.001

Table 24: Hard-bootstrap selector extension on Camelyon17 ERM ($\beta = 0.60$, ten seeds). Deltas are measured relative to the baseline-selected checkpoint. Proxy-only and val-loss-only are selectors on the same bootstrap runs; oracle loss is the best held-out-loss epoch within those same runs. The selector-level hazard is still the proxy-only versus safer-selector gap: proxy-only is slightly better on its own bootstrap proxy but materially worse on held-out loss, teacher-difficulty tail CVaR, and ECE. The fixed $1.25\times$ guardrail is included for completeness and is effectively slack on this family.

selected checkpoint's ordinary cross-entropy loss, then evaluate the proxy-selected checkpoint on that same slice. This does not identify a semantic subgroup, but it shows that the loss-tail movement is not only an artifact of the teacher-difficulty bank construction.

A.19 Early Warning Value of R_w

Table 27 and Figure 6 show that the same Camelyon17 ERM family-control stack already contains a practical early-warning signal. Using the rule "warn about harmful tail direction iff $R_w < 1$," early R_w computed well before final selection gives a strong same-stack warning about whether the eventual selected checkpoint will inflate or reduce the tail on the tested stack.

A.20 End-to-End Objective-Family Support

Table 28 summarizes the selected-checkpoint end-to-end Camelyon17 controls behind the mechanism discussion. It is included to show that the same evaluation protocol yields a harmful suppressive family and comparison families with lower held-out loss and teacher-tail diagnostics.

Contrast	Epochs	ΔGCE proxy	ΔAcc	ΔLoss	ΔECE	ΔTail	R_w
GCE proxy-best − ERM baseline-selected	14.4 / 3.8	-0.0243	+0.0039	+0.0449 [+0.0391,+0.0514]	+0.0274 [+0.0261,+0.0287]	+5.84 [+5.54,+6.17]	0.339
GCE proxy-best − GCE val-loss-best	14.4 / 3.3	-0.0146	+0.0098	+0.0181 [+0.0136,+0.0233]	+0.0129 [+0.0104,+0.0153]	+3.05 [+2.50,+3.60]	0.339
GCE fixed epoch 30 − ERM fixed epoch 30	30 / 30	-0.0173	+0.0068	+0.0648 [+0.0636,+0.0662]	+0.0308 [+0.0305,+0.0311]	+6.15 [+6.08,+6.21]	0.308

Table 25: Generalized cross-entropy (GCE, $q = 0.7$) extension on Camelyon17 ERM, ten seeds. Negative ΔGCE proxy means the GCE objective-aligned validation score improves. The same-trajectory comparison is GCE proxy-best minus GCE validation-loss-best; all three contrasts worsen held-out loss, ECE, and teacher-tail in $10/10$ seeds. Epochs report mean selected epoch for the first and second policy in each contrast.

Objective	ΔFixed loss-tail	ΔHeld-out loss	Harmful seeds
SoftClip P95	+0.942 [+0.836,+1.048]	+0.055 [+0.043,+0.068]	10/10
GCE q=0.7	+0.896 [+0.832,+0.959]	+0.045 [+0.037,+0.052]	10/10

Table 26: Teacher-free held-out loss-tail check on Camelyon17 ERM. The fixed loss-tail is the top decile of hospital-2 examples by ordinary cross-entropy under the corresponding baseline-selected checkpoint; the same fixed slice is then evaluated under the proxy-selected suppressive checkpoint. Intervals are paired 95% bootstrap intervals over ten seeds.

Early window	Seed sign acc.	Regime sign acc.	Regime Spearman
Epochs 1–3	0.97	1.00	-0.964
Epochs 1–5	0.97	1.00	-0.952
Epochs 1–10	0.97	1.00	-0.952

Table 27: Early R_w as a same-stack warning for final selected-checkpoint tail direction on the Camelyon17 ERM objective-family stack. "Regime" aggregates the ten-seed mean for each family setting, and the sign rule warns about harmful tail direction when $R_w < 1$. The warning is stable across short early windows.

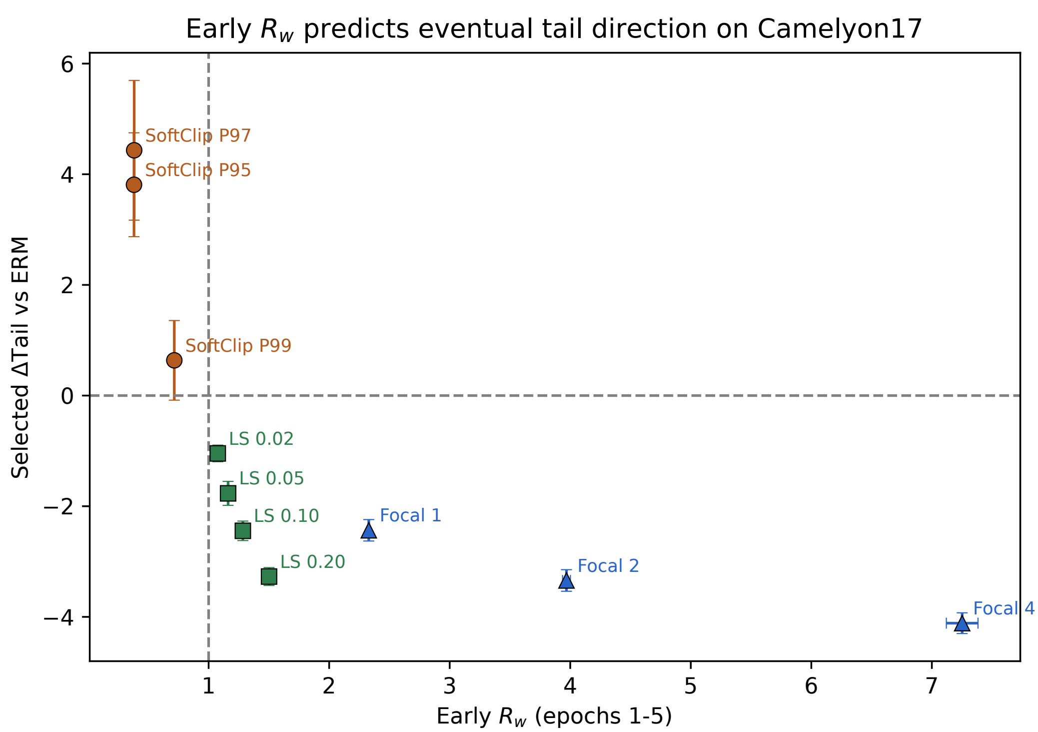

Figure 6: Early R_w as a same-stack warning for final selected ΔTail on the Camelyon17 ERM objective-family stack. Each point is a ten-seed regime mean with 95% mean confidence bars. The horizontal and vertical reference lines mark zero tail change and $R_w = 1$ respectively.

Family	ΔHeld-out loss$\downarrow$	ΔHeld-out acc$\uparrow$	ΔTail CVaR$\downarrow$	ΔECE$\downarrow$
SoftClip P95	+0.221	+0.047	+6.626	+0.025
Label smoothing 0.10	-0.276	+0.029	-2.647	+0.014
Focal 2	-0.280	-0.026	-2.526	+0.024

Table 28: Selected-checkpoint end-to-end objective-family support on a separate Camelyon17 finetune-control sweep. Deltas are measured relative to that sweep's own baseline finetune checkpoint. This table supports the method-ranking comparison in Table 3 and is not numerically comparable to the main selector-anchor finetune sweep in Tables 1 and 2. Soft clipping is the harmful suppressive family; label smoothing is the cleanest counterexample; focal is mixed.

B Broader Impact

The positive impact of this work is practical reliability. Many ML pipelines still report accuracy as the main model-selection signal, even when training objectives are already heavily shaped. Our results show that this can hide meaningful degradation in held-out loss, calibration, and group- or tail-sensitive behavior, including under end-to-end finetuning.

The main misuse risk is the opposite lesson: under suppressive objectives, proxy-aligned checkpoint selection can sometimes improve headline metrics while concealing reliability regressions. That makes it easier to overstate progress if only proxy-aligned metrics are reported. We therefore emphasize conservative reporting, explicit held-out-loss reporting and pre-specified safety-budget cross-checks, and lightweight diagnostic vetoes rather than proposing a stronger distortion-based selector.

C Reproducibility

All main reported sweeps use explicit seed sets and artifact-backed CSV summaries stored in the repository. The command-backed paper-facing tables and figures listed in the supplement are generated from those artifacts by scripts in `src/scripts`; a small number of supporting sensitivity/audit tables are included as static summaries with their status documented in the supplement. The submission package includes an anonymized supplement archive; reviewers should find `supplement/README.md`, `supplement/SUPPLEMENT_COMMANDS.md`, and `supplement/ASSET_TERMS.md` there.

The main training families use $n = 10$ seeds. Core Camelyon17 ERM runs train a linear head on cached ResNet50 embeddings for 30 epochs with AdamW, learning rate 10^{-3}, batch size 512, weight decay 10^{-4}, and evaluation batch size 2048. CelebA ERM uses cached ResNet50 embeddings with a width-256 head, AdamW, learning rate 10^{-3}, batch size 512, weight decay 10^{-4}, 30 epochs, and evaluation batch size 2048. Frozen CivilComments uses cached `distilbert-base-uncased` embeddings with a linear head, AdamW, learning rate 10^{-3}, batch size 512, weight decay 10^{-4}, 30 epochs, and evaluation batch size 2048. Camelyon17 finetuning uses a 10-epoch end-to-end recipe with AdamW, learning rate 10^{-4}, batch size 128, weight decay 10^{-4}, evaluation batch size 256, and `layer4` unfrozen. The appendix end-to-end CivilComments support uses `distilbert-base-uncased` finetuning for 10 epochs with AdamW, learning rate 2×10^{-5}, batch size 16, evaluation batch size 32, weight decay 0.01, and gradient clipping at 1.0. ACSIncome uses a width-256 two-hidden-layer MLP with Adam, learning rate 10^{-3}, batch size 2048, weight decay 10^{-4}, and 50 epochs. All reported training and evaluation were run on a single NVIDIA GeForce RTX 5090 workstation with 32607 MiB ($\approx$32 GB) VRAM; the longest saved main-supporting run in the repository is the ten-seed end-to-end CivilComments sweep, whose manifest records roughly 12 GPU-hours total. Camelyon17 and CivilComments are used through WILDS [18]; the underlying dataset sources include CAMELYON17 [3], CelebA [23], and Folktables/ACSIncome [9]. The supplement bundle includes the exact commands, minimal dependency list, and saved claim-bearing artifacts needed to regenerate the listed command-backed paper-facing outputs from saved repository artifacts, including the ACSIncome regression extension under `artifacts/metrics/acs_income/`. In particular, the bundled regeneration path covers the main selector table, the core-case summary, the Camelyon selector-diagnostic table, the Camelyon17 ERM selector-dose, validation-accuracy selector sensitivity, selector-averaging sensitivity, and leave-one-hospital-out selector-only diagnostics, the hard-bootstrap support table, the GCE $q = 0.7$ ERM corroboration table, the end-to-end GCE $q = 0.7$ finetune selector audit, the teacher-free held-out loss-tail table, the objective-family finetune support table, the persistence-support table, the per-bank robustness tables, the seed-count and paired-interval summaries, and the Camelyon dose-response appendix table from saved CSV artifacts rather than retraining. Static support tables and their provenance status are documented separately in the supplement. Viewed narrowly, the package therefore provides an artifact-backed regeneration path for the listed claim-bearing results and a compact selector-comparison suite for comparing alternative checkpoint selectors on the released runs without retraining. Full end-to-end reruns additionally require the external benchmark assets and the surrounding project environment. `supplement/ASSET_TERMS.md` summarizes the credited third-party assets and the verified governing terms used in the submission package. Camelyon17 is used through the WILDS patch-based release stored locally as `camelyon17_v1.0`. The current WILDS dataset page states that the public release is distributed under CC0, while the CAMELYON17 challenge website contains mixed license notices across its public pages. We therefore do not redistribute any raw third-party images in the supplement and restrict the artifact package to code and derived summaries.

We intentionally keep the main paper focused on a small set of claim-bearing outputs: the core-case summary, the selector-comparison table, the divergence figure, the ranking-reversal table, and the main same-stack mechanism exhibit.

NeurIPS Paper Checklist

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   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction scope the paper to suppressive-workflow hazards under proxy-selected reporting, distinguish baseline-referenced workflow comparisons from within-trajectory selector/admissibility analyses, include the narrower CivilComments and ACSIncome support cases, and state the end-to-end Camelyon17 result without presenting it as selector-only causal evidence.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: The paper explicitly covers anchor dependence, the asymmetric cross-modal scope of the evidence (frozen-embedding text versus end-to-end vision), the heuristic nature of the guardrail, the stress-test status of the main P95 regime, and the fact that persistence is regime-dependent rather than universal.

3. **Theory assumptions and proofs**

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5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

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Answer: [Yes]

Justification: The paper states the datasets, deployment splits, training families, P95 distortion setting, guardrail budgets, selected-versus-fixed protocol, seed counts, and evaluation metrics, and the reproducibility appendix lists the main optimizer, learning-rate, batch-size, epoch, and finetuning details used by the claim-bearing runs.

7. **Experiment statistical significance**

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: The main text surfaces paired bootstrap intervals for the principal loss / reliability / ECE harms on the three seed-matched hazard anchors, the appendix reports paired intervals for the Camelyon same-trajectory selector, selector-dose, validation-accuracy selector-sensitivity diagnostics, and the change from proxy-only to the diagnostic veto, and the main tables are built from explicit seed-matched summaries. The leave-one-hospital-out selector-only table is reported as a fold-level directional robustness check.

8. **Experiments compute resources**

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: The reproducibility section states that the reported supervised experiments were run on a single NVIDIA GeForce RTX 5090 workstation, gives representative training hyperparameters, records the longest saved supporting run at roughly 12 GPU-hours total, and notes that the final paper assets are generated from saved repository artifacts by explicit scripts.

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Answer: [Yes]

Justification: The supplement now includes `ASSET_TERMS.md`, which credits the third-party datasets, benchmark suites, and code assets used in the supervised experiments, records the verified governing terms where available (including MIT, BSD-3-Clause, CC0, research-use terms, and public ACS microdata terms), and documents cases with ambiguous public notices. The submission redistributes only derived summaries and paper-facing artifacts, not raw images or benchmark shards.

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Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

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Justification: The paper introduces paper-specific evaluation artifacts for the teacher-difficulty tail diagnostic on the vision anchors, and those artifacts are documented in Appendix A.4 and the supplement through the bank-construction description, exact script names, paper-facing bankwise summaries, and artifact-backed regeneration commands. These are documented as reproducibility assets rather than as a standalone benchmark contribution.

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